

Mautic 5.x on Kubernetes

Deployment Guide & Checklist

Supplementary material for: [How to Set Up Mautic 5.x on Kubernetes: A Step-by-Step Guide](https://kozlov.ski/mautic-5x-kubernetes-setup/)
(<https://kozlov.ski/mautic-5x-kubernetes-setup/>)

Overview

This guide walks you through deploying Mautic 5.x on a local Kubernetes cluster using Minikube. By the end, you'll have a fully functional Mautic instance with MariaDB for data persistence and Mailhog for email testing.

Estimated completion time: 30–45 minutes

Checkpoint 0: Prerequisites

System Requirements

Ensure your system meets the following requirements before proceeding:

- ☐ Docker installed and running
- ☐ At least 4GB RAM available for Minikube
- ☐ At least 20GB free disk space

Install Required Tools

Install the Kubernetes CLI tools using Homebrew:

```
brew install kubectl  
brew install minikube  
brew install helm
```

Verify Installation

- ☐ Verify kubectl: `kubectl version --client`
- ☐ Verify minikube: `minikube version`
- ☐ Verify helm: `helm version`

Start Minikube Cluster

```
minikube start --driver=docker
```

Enable Required Addons

```
minikube addons enable metrics-server  
minikube addons enable ingress
```

Verification checklist:

- ☐ Minikube cluster is running (`minikube status`)
- ☐ Metrics server addon is enabled
- ☐ Ingress addon is enabled

Checkpoint 1: Configure Helm Repositories

Add the required Helm chart repositories:

```
helm repo add bitnami https://charts.bitnami.com/bitnami  
helm repo add codecentric https://codecentric.github.io/helm-charts  
helm repo add mautic https://mautic.github.io/helm-charts  
helm repo update
```

Verification checklist:

- ☐ Bitnami repo added successfully
- ☐ Codecentric repo added successfully
- ☐ Mautic repo added successfully
- ☐ All repos updated (`helm repo list` shows all three)

Checkpoint 2: Deploy MariaDB

MariaDB serves as the database backend for Mautic.

Install MariaDB

```
helm install mariadb bitnami/mariadb -f mariadb-values.yaml
```

Optional: External Database Access

To connect to the database from outside the cluster (useful for debugging):

```
kubectl apply -f mariadb-nodeport-service.yaml  
minikube service mariadb-nodeport --url
```

Verification checklist:

- ☐ MariaDB pod is running (`kubectl get pods | grep mariadb`)
- ☐ MariaDB service is available (`kubectl get svc | grep mariadb`)

Checkpoint 3: Deploy Mailhog

Mailhog captures outgoing emails for testing purposes.

Install Mailhog

```
helm install mailhog codecentric/mailhog -f mailhog-values.yaml
```

Verification checklist:

- ☐ Mailhog pod is running (`kubectl get pods | grep mailhog`)
- ☐ Mailhog service is available (`kubectl get svc | grep mailhog`)

Checkpoint 4: Deploy Mautic

Step 4.1: Create Custom Favicon ConfigMap

Before installing the Helm chart, create the required Kubernetes ConfigMap for the custom favicon:

```
kubectl apply -f ct-favicon-configmap.yaml
```

☐ ConfigMap created successfully

Step 4.2: Configure MaxMind IP Lookup (Optional)

Mautic uses MaxMind GeoLite2 for IP geolocation to identify visitor locations. To enable this feature:

1. Create a free MaxMind account at [maxmind.com/en/geolite2/signup](https://www.maxmind.com/en/geolite2/signup) (<https://www.maxmind.com/en/geolite2/signup>)
2. Generate a license key in your account dashboard under **Manage License Keys**
3. Update `mautic-values.yaml` with your credentials:

```
MAUTIC_IP_LOOKUP_AUTH: "<ACCOUNT_ID>:<LICENSE_KEY>"
```

Note: This step is optional. Mautic will function without IP lookup, but you won't have geographic data for your contacts.

- ☐ MaxMind account created (optional)
- ☐ License key configured in values file (optional)

Step 4.3: Install Mautic via Helm

```
helm install mautic mautic/mautic -f mautic-values.yaml --version 5.1.71
```

Useful commands for managing Mautic:

```
# Upgrade existing installation
helm upgrade mautic mautic/mautic --version 5.1.71 --dry-run
```

```
# Remove installation completely
helm delete mautic
```

☐ Mautic Helm release installed

Step 4.4: Resolve Health-Check Issues

After installation, clear the cache and fix permissions to resolve potential 500 errors:

```
# Clear application cache
kubectl exec -it $(kubectl get pod -l app.kubernetes.io/name=mautic -o
  jsonpath='{.items[0].metadata.name}') \
  -- php bin/console cache:clear

# Sync database migrations metadata
kubectl exec -it $(kubectl get pod -l app.kubernetes.io/name=mautic -o
  jsonpath='{.items[0].metadata.name}') \
  -- php bin/console doctrine:migrations:sync-metadata-storage

# Fix file ownership
kubectl exec -it $(kubectl get pod -l app.kubernetes.io/name=mautic -o
  jsonpath='{.items[0].metadata.name}') \
  -- chown -R www-data:www-data var
```

Verification checklist:

- ☐ Cache cleared successfully
- ☐ Migrations metadata synced
- ☐ File ownership corrected
- ☐ Mautic pod is running without restarts (`kubectl get pods | grep mautic`)

Checkpoint 5: Access Services

Enable Network Tunnel

Start the Minikube tunnel to expose services locally (requires sudo):

```
sudo minikube tunnel
```

Note: Keep this terminal window open while accessing the services.

Access URLs

SERVICE	URL	CREDENTIALS
Mautic	http://mautic.local	admin / Maut1cR0cks!
Mailhog	http://mailhog.local	—

Verification checklist:

- ☐ Minikube tunnel is running
- ☐ Mailhog web interface is accessible
- ☐ Mautic login page loads
- ☐ Successfully logged into Mautic admin panel

Checkpoint 6: Verify API Access (Optional)

Test the Mautic API to ensure everything is working correctly.

Generate Authorization Header

```
echo "Authorization: Basic $(echo -n "admin:Maut1cR0cks!" | base64)"
```

Test API Endpoints

Create a new stage:

```
curl -X POST \
  'http://mautic.local/api/stages/new' \
  -H "Authorization: Basic YWRtaW46TWFlbDFjUjBja3Mh" \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Top of Funnel",
    "description": "Stage for newly acquired customers",
```

```
"weight": 1,  
"isPublished": true  
}'
```

Create a new segment:

```
curl -X POST \  
  'http://mautic.local/api/segments/new' \  
  -H "Authorization: Basic YWRtaW46TWF1dDFjUjBja3Mh" \  
  -H 'Content-Type: application/json' \  
  -d '{  
    "name": "Top of Funnel",  
    "description": "Top of Funnel Users",  
    "isPublished": 1,  
    "filters": [  
      {  
        "glue": "and",  
        "object": "lead",  
        "field": "stage",  
        "type": "stage",  
        "operator": "=",  
        "properties": {"filter": 1}  
      }  
    ]  
  }'
```

Create a new contact:

```
curl -X POST \  
  'http://mautic.local/api/contacts/new' \  
  -H "Authorization: Basic YWRtaW46TWF1dDFjUjBja3Mh" \  
  -H 'Content-Type: application/json' \  
  -d '{  
    "firstname": "John",  
    "lastname": "Doe",  
    "email": "john.doe@example.com",  
    "points": 10  
  }'
```

Verification checklist:

- ☐ Stage created successfully
- ☐ Segment created successfully
- ☐ Contact created successfully
- ☐ Test email sent to contact appears in Mailhog

Checkpoint 7: Cleanup

When you're finished, clean up the local environment:

```
minikube stop  
minikube delete
```

- ☐ Minikube stopped
- ☐ Minikube cluster deleted

Appendix: Troubleshooting

Manual Configuration Updates

To manually update the Mautic configuration in the `config/local.php` file:

1. Access the Mautic pod:

```
kubectl exec -it $(kubectl get pod -l app.kubernetes.io/name=mautic -o  
jsonpath='{.items[0].metadata.name}') -- /bin/bash
```

2. Fix file permissions:

```
chmod 664 config/local.php  
chown www-data:www-data config/local.php
```

3. Edit the configuration file as needed, then update `mautic-values.yaml` accordingly.

Common Issues

ISSUE	SOLUTION
Mautic returns 500 error	Run cache clear and permissions fix commands from Checkpoint 4.4
Cannot access *.local domains	Ensure <code>minikube tunnel</code> is running with sudo
Pods stuck in pending state	Check available resources with <code>kubectl describe pod <pod-name></code>

Quick Reference

Essential Commands

```
# Check pod status
kubectl get pods

# View pod logs
kubectl logs <pod-name>

# Access pod shell
kubectl exec -it <pod-name> -- /bin/bash

# List all services
kubectl get svc

# Check Helm releases
helm list
```

Questions or issues? Refer to the [full blog post \(https://kozlov.ski/mautic-5x-kubernetes-setup/\)](https://kozlov.ski/mautic-5x-kubernetes-setup/) for detailed explanations and troubleshooting tips.